

Self-Learning: Topological Sorting -- Data Structures

Definition: Topological Sorting of a Directed Acyclic Graph (DAG) is a linear ordering of vertices such that for every directed edge (u, v), vertex u comes strictly before vertex v in the ordering.

1. Detailed Technical Explanation

Sample DAG (Task Dependencies):

(5)

(0)

<----

(4)

|

|

v

v

(2)

(3)

(1)

Valid Topological Orders:

- 5 -> 4 -> 2 -> 3 -> 1 -> 0

- 4 -> 5 -> 2 -> 3 -> 1 -> 0

- 5 -> 2 -> 3 -> 4 -> 1 -> 0

Key Conditions:

1. Topological sorting is defined ONLY for DAGs (Directed Acyclic Graphs).

2. Algorithms for Topological Sorting

1. Kahn's Algorithm (BFS In-Degree Approach)

1. Compute the In-Degree of every vertex in the graph.

2. DFS with Stack Approach

1. Initialize a visited[] boolean array.

3. Real-World Applications

- * Build Systems (Make, Gradle, Webpack): Resolving source code compilation dependencies.

4. Quick Recall Flow

DAG Only -> In-Degree = 0 Enqueue -> Decrement Neighbors -> Repeat -> Linear Dependency Order (Kahn's Algorithm)